The Convex Origin of Fixed Costs[1]
Online Appendix

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A Proof of Proposition 4 (Multi-action duality)

Proof.
Part (i): General case. The agent observes cost draws $(\xi_1, \ldots, \xi_J)$ with $\xi_j \sim G_j$ independently, and selects the action with the highest net payoff:

$$\Pi^{FC}(\mathbf{S}) = E\left[\max_{j\in\{0,\ldots,J\}} \{S_j - w\xi_j\}\right].$$

Partition the expectation by which action is optimal. For each $j \geq 1$, action j is chosen when $S_j - w\xi_j \geq S_k - w\xi_k$ for all $k \neq j$ and $S_j - w\xi_j \geq 0$. Under independence, the probability that action j is chosen can be written as a function $\varphi_j(\mathbf{S})$, and the expected cost paid conditional on choosing j is $E[w\xi_j \mid j \text{ chosen}]$.

Define $\varphi_0 = 1 - \sum_{j=1}^{J} \varphi_j$ (the probability of inaction). The expected payoff is:

$$\Pi^{FC}(\mathbf{S}) = \sum_{j=0}^{J} \varphi_j^* S_j - \sum_{j=1}^{J} E[w\xi_j \cdot \mathbf{1}_{j \text{ chosen}}].$$

By the Fenchel conjugate construction, $\Pi^{FC}(\mathbf{S}) = \sup_{\boldsymbol{\varphi}\in\Delta^J}\{\sum_j \varphi_j S_j - C(\boldsymbol{\varphi})\}$ where $C(\boldsymbol{\varphi})$ is the convex conjugate of Π^{FC}. Under independence of the G_j, the cost function inherits a separable structure: the marginal cost of raising φ_j depends only on G_j, not on the other distributions.

Joint convexity of C follows from the fact that $\Pi^{FC}(\mathbf{S})$, as a pointwise maximum of affine functions of $\mathbf{S}$, is convex in $\mathbf{S}$. The Fenchel biconjugate theorem then implies that $C = (\Pi^{FC})^*$ is convex on Δ^J.

Part (ii): i.i.d. Gumbel specialisation. Let $\xi_j \overset{i.i.d.}{\sim} \text{Gumbel}(\mu, \sigma)$ for $j = 0, 1, \ldots, J$, with CDF $G(\xi) = e^{-e^{-(\xi-\mu)/\sigma}}$. The expected maximum of $\{S_j - w\xi_j\}$ is a classical result (McFadden, 1974):

$$\Pi^{FC}(\mathbf{S}) = \sigma \ln\left(\sum_{j=0}^{J} e^{S_j/\sigma}\right) + \mu',$$

where $\mu' = w\mu - w\sigma\gamma_E$ and γ_E is the Euler–Mascheroni constant. This is the log-sum-exp function (up to constants), which is convex in $\mathbf{S}$.

The gradient gives the choice probabilities:

$$\varphi_j^* = \frac{\partial \Pi^{FC}}{\partial S_j} = \frac{e^{S_j/\sigma}}{\sum_{k=0}^{J} e^{S_k/\sigma}},$$

the multinomial logit. The Fenchel conjugate of the log-sum-exp is the negative entropy:

$$C(\boldsymbol{\varphi}) = \sup_{\mathbf{s}} \left\{ \sum_j \varphi_j S_j - \sigma \ln\left(\sum_j e^{S_j/\sigma} \right) \right\} - \mu' = \sigma \sum_{j=0}^{J} \varphi_j \ln \varphi_j,$$

which is non-negative on Δ^J (since $\ln \varphi_j \leq 0$) and jointly convex. ■

Remark 1. *The Gumbel distribution has unbounded support $(-\infty, \infty)$, so negative cost draws are possible. In the original fixed cost interpretation, this means that some realisations yield a "bonus" rather than a cost. The duality holds provided $E[\max_j\{S_j - w\xi_j\}]$ is finite, which is guaranteed by the double-exponential tails of the Gumbel. For distributions on $[0, \bar{\xi}]$ (bounded support), the convergence is automatic.*

B Proof of Corollary 5 ($\kappa = (1 + 2\gamma)\bar{\Lambda}$ for power laws)

Proof.
For $G(\xi) = (\xi/\bar{\xi})^\gamma$, the density is $g(\xi) = (\gamma/\bar{\xi})(\xi/\bar{\xi})^{\gamma-1}$. Evaluate the intensive margin integrand at $\xi = Bx^2/w$:

$$\begin{aligned} g\left(\frac{Bx^2}{w}\right) \cdot \frac{2Bx^2}{w} &= \frac{\gamma}{\bar{\xi}} \left(\frac{Bx^2/w}{\bar{\xi}}\right)^{\gamma-1} \cdot \frac{2Bx^2}{w} \\ &= 2\gamma \cdot \frac{1}{\bar{\xi}} \left(\frac{Bx^2/w}{\bar{\xi}}\right)^{\gamma-1} \cdot \frac{Bx^2}{w} \\ &= 2\gamma \left(\frac{Bx^2}{w\bar{\xi}}\right)^{\gamma} = 2\gamma\, \Lambda(x). \end{aligned} \tag{1}$$

Substituting into equation (20) of the main paper:

$$\begin{aligned} \kappa(G) &= \int G\left(\frac{Bx^2}{w}\right) f_{ss}(x)\, dx + \int 2\gamma\, \Lambda(x)\, f_{ss}(x)\, dx \\ &= \bar{\Lambda} + 2\gamma\bar{\Lambda} = (1 + 2\gamma)\bar{\Lambda}. \end{aligned}$$

The selection wedge is $\kappa - \bar{\Lambda} = 2\gamma\bar{\Lambda}$. The ratio $\kappa/\bar{\Lambda} = 1 + 2\gamma$ depends only on γ, not on f_{ss}, B, or w. ■

C Proof of Corollary 4 (Ordering of non-neutrality)

Proof.
The real effect of a monetary shock dm is proportional to $(1-\kappa)\,dm$, so non-neutrality is decreasing in κ.

(i) Calvo benchmark. When Λ is constant at λ, the intensive margin vanishes: $g(Bx^2/w)\cdot 2Bx^2/w = 0$ for all x in the Calvo limit, since a constant hazard implies $g = 0$ almost everywhere. Therefore $\kappa = \bar{\Lambda} = \lambda$ and the selection wedge is zero. Non-neutrality is $1-\lambda$, the maximum among models with the same frequency λ.

(ii) Absolutely continuous G with positive density. By Proposition 1(iii), $g > 0$ implies $C'' > 0$, so $\Lambda(x)$ is strictly increasing in $|x|$. The intensive margin $\int g(Bx^2/w)(2Bx^2/w)f_{ss}(x)\,dx > 0$ since the integrand is non-negative and strictly positive on a set of positive f_{ss}-measure. Therefore $\kappa > \bar{\Lambda}$ and the selection wedge is strictly positive.

(iii) Golosov-Lucas. When G is degenerate at $\bar{\xi}$, $\Lambda(x) = \mathbf{1}[Bx^2 \geq w\bar{\xi}]$. All adjusters have gaps at or beyond the threshold $\bar{x} = \sqrt{w\bar{\xi}/B}$, so the intensive margin is maximised (concentrated entirely at the boundary). Non-neutrality is minimal.

Monotonicity in γ: For the power law family, $\kappa = (1+2\gamma)\bar{\Lambda}$ by Corollary 5. At fixed $\bar{\Lambda}$, κ is strictly increasing in γ and non-neutrality $1-\kappa$ is strictly decreasing. ■

D Proof of Proposition 9 (Selection and the menu cost distribution)

Proof.
Write the adjustment-weighted mean absolute gap as

$$\overline{|x|}^{adj} = \frac{E[\Lambda(x)\cdot|x|]}{E[\Lambda(x)]} = E[|x|] + \frac{\mathrm{Cov}(\Lambda(x)/\bar{\Lambda},\ |x|)}{1}.$$

The covariance is positive when Λ is increasing in $|x|$. Its magnitude depends on how sharply Λ responds to $|x|$.

(i) The elasticity of the hazard with respect to $|x|$ is

$$\frac{\partial \ln \Lambda(x)}{\partial \ln |x|} = \frac{g(Bx^2/w)}{G(Bx^2/w)}\cdot\frac{2Bx^2}{w},$$

which is twice the hazard rate of G evaluated at the firm's surplus. A higher hazard rate means Λ responds more sharply to $|x|$, increasing the covariance between $\Lambda(x)/\bar{\Lambda}$ and $|x|$, and hence increasing $\mathcal{S}$.

(ii) By Proposition 1(iii), $C'' = w/g(G^{-1}(\varphi))$. When g is small at the threshold, C'' is large and Λ responds steeply to changes in x^2: few agents are clustered near the margin, so a small increase in the surplus tips few new firms into adjustment but those who do adjust have substantially larger gaps. This concentrates adjustment among high-gap firms, strengthening selection.

(iii) For constant Λ (Calvo), the covariance is zero and $\mathcal{S} = 0$. For a step function Λ (Golosov-Lucas), adjustment is deterministic above the threshold: $\mathrm{Cov}(\Lambda/\bar{\Lambda}, |x|)$ is maximal since $\Lambda/\bar{\Lambda}$ perfectly separates high-gap from low-gap firms.

For the power law family, the hazard rate of G at ξ is $g(\xi)/G(\xi) = \gamma/\xi$, so the elasticity is 2γ — constant and strictly increasing in γ. ■

E Proof of Proposition 6 (Investment response and the cost density)

Proof.
By Theorem 1, the optimal investment probability is $\varphi^*(k) = G(S(k)/w)$. Differentiating with respect to the surplus:

$$\frac{d\varphi^*}{dS} = \frac{g(S/w)}{w} = \frac{g(G^{-1}(\varphi^*))}{w} = \frac{1}{C''(\varphi^*)},$$

where the last equality uses $C''(\varphi) = w/g(G^{-1}(\varphi))$ from Proposition 1(iii). This is the density of cost draws at the current threshold, scaled by $1/w$: the number of firms on the margin of investing per unit surplus.

Taking expectations across the cross-section of firms:

$$\frac{dE[\varphi^*(k)]}{dS} = E\left[\frac{g(G^{-1}(\varphi^*(k)))}{w}\right],$$

which is the expected density at the threshold, averaged over the distribution of capital stocks k. This is positive and proportional to the density of agents at the adjustment margin. ■

F Proof of Proposition 8 (Stationary distribution)

Proof.
The price gap evolves according to:

$$x_{t+1} = \begin{cases} z_{t+1} & \text{with probability } \Lambda(x_t), \\ x_t + z_{t+1} & \text{with probability } 1 - \Lambda(x_t), \end{cases}$$

where $z_{t+1} \sim \phi_z$ independently. The density of x_{t+1} conditional on the current cross-section f_t is:

$$f_{t+1}(x) = \int \Lambda(\tilde{x}) f_t(\tilde{x})\, d\tilde{x} \cdot \phi_z(x) + \int (1 - \Lambda(\tilde{x}))\phi_z(x - \tilde{x}) f_t(\tilde{x})\, d\tilde{x}.$$

The first term: adjusting firms (total mass $\bar{\Lambda}_t$) draw $z_{t+1} \sim \phi_z$ as their new gap. The second term: non-adjusting firms at $\tilde{x}$ receive shock z_{t+1} and move to $\tilde{x} + z_{t+1}$; the density at x is the convolution of $(1 - \Lambda(\tilde{x}))f_t(\tilde{x})$ with ϕ_z.

Setting $f_{t+1} = f_t = f_{ss}$ yields the fixed-point equation (13) of the main paper. Symmetry: if ϕ_z is symmetric around zero and $\Lambda(x) = \Lambda(-x)$ (which holds since $\Lambda(x) = G(Bx^2/w)$), then if f_t is symmetric, so is f_{t+1}. Since f_{ss} is the unique fixed point of this contraction (the operator is a weighted average of a density and a convolution, both preserving the space of symmetric densities), f_{ss} is symmetric around zero. ■

G Comparison with Williams-Daly-Zachary

Section 2.2 of the main paper discusses the relationship between Theorem 1 and the Williams-Daly-Zachary (WDZ) theorem (McFadden, 1978, 1981). This appendix provides a systematic comparison. Both results share a common mathematical skeleton — an expected-maximum operator equals the Fenchel conjugate of a function determined by a distributional primitive — but differ in their economic content, scope, and implications. Table 1 summarises the comparison along six dimensions.

For binary choice ($J = 2$), the two results are formally equivalent under a sign change: a random benefit $S - w\xi$ versus a random cost $w\xi$ applied to a fixed benefit S are isomorphic. The binary conjugate identity is therefore not new. The contributions of this paper lie beyond the duality identity itself: the mapping $G \mapsto C$ and the distributional taxonomy (Propositions 3–5), the structural explanation for the Khan and Thomas (2008) result, and the continuous parametrisation of the pricing spectrum are entirely absent from WDZ and the discrete choice literature.

Table 1: Williams-Daly-Zachary vs. Theorem 1

Dimension	**WDZ**	**Theorem 1**
Random element	Taste shocks ε_j on payoffs	Cost draws $w\xi$ on adjustment
Decision	Multi-alternative discrete choice	Binary: adjust or not
Surplus function	$E[\max_j\{u_j + \varepsilon_j\}]$	$E[\max\{S - w\xi,\ 0\}]$
Convexity result	Surplus convex in $(u_1, \ldots, u_J)$	Payoff convex in S
Conjugate of	Generating function $\mathcal{G}(\varepsilon)$	$C(\varphi) = \int_0^\varphi G^{-1}(t)\,dt$
Choice probs	$p_j = \partial E[\max]/\partial u_j$	$\varphi^* = G(S)$
Primitive	Joint distribution of ε	Distribution G of costs
Coupling	Not constructed	Path-by-path equivalence
Economic context	Demand: consumer choice	Supply: adjustment frictions
Application	Multinomial logit (McFadden, 1974)	Calvo–GL spectrum

References

Khan, A. and J. K. Thomas (2008): "Idiosyncratic Shocks and the Role of Nonconvexities in Plant and Aggregate Investment Dynamics," *Econometrica*, 76, 395–436.

McFadden, D. (1974): "Conditional Logit Analysis of Qualitative Choice Behavior," in *Frontiers in Econometrics*, ed. by P. Zarembka, Academic Press, 105–142.

——— (1978): "Modelling the Choice of Residential Location," in *Spatial Interaction Theory and Planning Models*, ed. by A. Karlqvist et al., North-Holland, 75–96.

——— (1981): "Econometric Models of Probabilistic Choice," in *Structural Analysis of Discrete Data*, ed. by C. F. Manski and D. McFadden, MIT Press, 198–272.